TAUTNESS IS PRESERVED BY BASE CHANGE OF LINEARLY TOPOLOGIZED RINGS

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Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file formal-spaces.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The first sentence of the proof refers to an open ideal of A whose image in C is contained in J, while the ideal introduced in the preceding sentence is called K; the text is reproduced verbatim as delivered and no correction was inserted.

Editorial note. The counted spans carry 1 navigation stub; no bibliography citation occurs in the spans.

Editorial note. The preamble span points at the chapter's own preamble.tex shipped with the frozen source, used verbatim as staged.

Allowed standard prerequisites (editorial).

- The completed tensor product as the limit of the tensor products of the quotients, and its fundamental system of open ideals. Same chapter, the description of the completed tensor product used at proof lines 1397-1400. Identifies the ideals generated by the chosen ideals as a fundamental system of open ideals.
- Tautness of a continuous map of linearly topologized rings. Same chapter, the definition of a taut map, in force in the statement. Fixes what has to be shown for the map from C.

Lemma 0.1. *Let $A \to B$ and $A \to C$ be continuous homomorphisms of linearly topologized rings. If $A \to B$ is taut, then $C \to B\widehat{\otimes}_A C$ is taut.*

Proof. Let $K \subset C$ be an open ideal. Choose any open ideal $I \subset A$ whose image in C is contained in J. By assumption the closure J of IB is open. Since $A \to B$ is taut we see that $B\widehat{\otimes}_A C$ is the limit of the rings $B/J \otimes_{A/I} C/K$ over all choices of K and I, i.e, the ideals $J(B\widehat{\otimes}_A C) + K(B\widehat{\otimes}_A C)$ form a fundamental system of open ideals. Now, since $B \to B\widehat{\otimes}_A C$ is continuous we see that J maps into the closure of $K(B\widehat{\otimes}_A C)$ (as I maps into K). Hence this closure is equal to $J(B\widehat{\otimes}_A C) + K(B\widehat{\otimes}_A C)$ and the proof is complete. □

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